Bernoulli Maximum-Likelihood Estimation, Part I: The Theoretical Solution and the Estimator's Report Card

Yuval Lavie

August 11, 2026

Abstract

Pure statistics — no losses, no descent. From i.i.d. samples of a Bernoulli distribution we derive the maximum-likelihood estimator in closed form, then judge it the way a statistician judges any estimator: bias, variance, efficiency against the Cramér–Rao bound, mean squared error, consistency, asymptotic normality (hence confidence intervals), and sufficiency. The companion script `mle_theoretical.py` verifies the closed form against a brute-force search and every quantitative property by Monte Carlo over thousands of replicated experiments. The translation of this problem into a machine-learning loss, and its solution by gradient methods, is Part II (`mle_practical.tex`).

1 From the likelihood to the estimator

Let $x_1, \ldots, x_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$ with $x_i \in \{0, 1\}$ and unknown $p \in (0, 1)$. The question of maximum likelihood: which p makes the observed sample most probable?

Step 1: the likelihood is a product of two kinds of factors. By independence, the probability of observing exactly this sample is

$$L(p) \;=\; \prod_{i=1}^{n} \Pr(x_i) \;=\; \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i}. \tag{1}$$

Each factor is just a case split written as one expression: it equals p when $x_i = 1$ and $1-p$ when $x_i = 0$. So if the sample contains $m = \sum_i x_i$ ones and $n - m$ zeros, collecting the factors gives

$$L(p) \;=\; p^{m}\,(1-p)^{n-m}. \tag{2}$$

Already here the data enters only through the count m: how the ones and zeros are ordered is irrelevant. (This observation becomes *sufficiency* in Section 2.)

Step 2: the logarithm turns the product into a sum. log is strictly increasing, so it preserves the location of a maximum: $\arg\max_p L(p) = \arg\max_p \log L(p)$. Applying it to (2),

$$\ell(p) \;=\; \log L(p) \;=\; m \log p + (n-m)\log(1-p). \tag{3}$$

This is the whole reason for working in logs: powers become products, products become sums, and sums are easy to differentiate (and do not underflow numerically the way a product of 5000 numbers in $(0, 1)$ would).

The closed form.

Proposition 1. *Assume* $0 < m < n$*. Then* $\hat{p}_{\text{MLE}} = m/n = \bar{x}$*, and it is the unique maximizer of* ℓ *on* $(0,1)$*.*

Proof. Differentiate (3) term by term:

$$\ell'(p) \;=\; \frac{m}{p} - \frac{n-m}{1-p}.$$

Set it to zero and clear denominators:

$$\frac{m}{p} = \frac{n-m}{1-p} \quad\Longrightarrow\quad m(1-p) = (n-m)\,p \quad\Longrightarrow\quad m - mp = np - mp \quad\Longrightarrow\quad m = np,$$

hence

$$\boxed{\hat{p}_{\text{MLE}} \;=\; \frac{m}{n} \;=\; \bar{x}} \tag{4}$$

— the fraction of ones in the sample. Uniqueness: $\ell''(p) = -m/p^2 - (n-m)/(1-p)^2 < 0$, so ℓ is strictly concave on $(0,1)$ and the stationary point is the global maximum. □

Remark 1 (Edge cases)**.** *If* $m = 0$ *or* $m = n$*, every factor of the likelihood pulls in the same direction:* L *is monotone on* $(0,1)$ *and its supremum sits at the boundary, which is again* m/n *— the formula survives.*

2 The estimator's report card

An estimator is a *random variable*: a new sample produces a new $\hat{p}$. So $\hat{p}$ has a distribution of its own, and we judge it by that distribution's shape. Throughout, use $m \sim \text{Binomial}(n,p)$, so $\mathbb{E}[m] = np$ and $\text{Var}(m) = np(1-p)$.

Bias. $\mathbb{E}[\hat{p}] = \mathbb{E}[m]/n = p$: the estimator is *unbiased* — over repeated samples it is centered on the truth exactly, at every n.

Variance and standard error.

$$\text{Var}(\hat{p}) \;=\; \frac{\text{Var}(m)}{n^2} \;=\; \frac{p(1-p)}{n}, \qquad \text{SE} \;=\; \sqrt{\frac{p(1-p)}{n}}, \tag{5}$$

estimated in practice by plugging $\hat{p}$ in for p. The $1/n$ decay is the rate at which data buys precision.

Efficiency: the Cramér–Rao bound is attained. The Fisher information of a single Bernoulli observation is

$$I(p) \;=\; \mathbb{E}\!\left[-\frac{d^2}{dp^2}\log f(X;p)\right] \;=\; \mathbb{E}\!\left[\frac{X}{p^2} + \frac{1-X}{(1-p)^2}\right] \;=\; \frac{1}{p} + \frac{1}{1-p} \;=\; \frac{1}{p(1-p)},$$

and the Cramér–Rao inequality states that *every* unbiased estimator T of p obeys $\text{Var}(T) \geq 1/\big(nI(p)\big)$ (derivation: `Theory/statistics`). Here

$$\frac{1}{n\,I(p)} \;=\; \frac{p(1-p)}{n} \;=\; \text{Var}(\hat{p})\,: \tag{6}$$

the bound holds with *equality*. The MLE is *efficient* — provably unimprovable among unbiased estimators, at every finite n.

Mean squared error. The general decomposition $\mathrm{MSE} = \mathrm{Var} + \mathrm{Bias}^2$ collapses to $\mathrm{MSE}(\hat{p}) = p(1-p)/n$: for unbiased estimators, accuracy is purely a variance story.

Consistency. Unbiasedness plus vanishing variance forces $\hat{p} \to p$ in probability (Chebyshev). Since $\hat{p} = \bar{x}$, this *is* the law of large numbers, specialized to coin flips.

Asymptotic normality, and what it buys. By the central limit theorem,

$$\sqrt{n}\,(\hat{p} - p) \ \xrightarrow{d}\ \mathcal{N}\big(0,\ p(1-p)\big), \tag{7}$$

so for large n the sampling distribution of $\hat{p}$ is approximately $\mathcal{N}(p,\, p(1-p)/n)$. This is what buys *confidence intervals*:

$$\hat{p}\ \pm\ 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \tag{8}$$

covers the true p in about 95% of repeated experiments. (The asymptotic variance $1/I(p)$ appearing here is no accident — it is the general asymptotic behavior of MLEs.)

Sufficiency. The likelihood (2) depends on the data only through m — the factorization criterion — so m is a *sufficient statistic*: compressing n observations into one count loses nothing about p.

Remark 2 (Invariance, used in Part II)**.** *The MLE of any function $g(p)$ is $g(\hat{p})$. This is why Part II may reparameterize the problem through a sigmoid and still recover the same estimator.*

3 Experiment

With $p = 0.3$ and $n = 5000$ (seed 7): $m = 1489$, so $\hat{p} = 0.297800$, agreeing with a brute-force maximization of (3) over a 10^5-point grid. The report card, verified by Monte Carlo over $R = 5000$ replicated experiments: empirical mean of $\hat{p}$ is 0.299910 against $p = 0.3$ (unbiasedness); empirical variance 4.244×10^{-5} against $p(1-p)/n = 4.200 \times 10^{-5}$, which equals the Cramér–Rao bound (efficiency); the 95% interval (8) is $[0.2851, 0.3105]$ and contains the truth. The figure shows, left to right: the true and fitted probability mass functions side by side; the log-likelihood (3) with its maximum at (4); and the sampling distribution of $\hat{p}$ — the histogram of the 5000 replicated estimates — with the CLT normal (7) drawn over it, indistinguishable.

One sentence to keep: the estimator is a random variable, and everything we proved — unbiased, efficient, normal — is a statement about *its* distribution, not about any single number.